Expert Validation Pre-Read

Ethical Concern Clusters and Operational Action Catalogue

Purpose of the study

This study evaluates a proposed solution for organizing ethical concern clusters and translating them into operational actions for software and digital systems. We are evaluating the research artifact - not your organization, its maturity, or its current practices.

1. Where the concern clusters come from

Evidence base. We identified **94 ethical concerns** from two complementary sources: (i) a survey with industry professionals and (ii) an academic systematic mapping study. The concerns were summarized and consolidated into 24 concern clusters so they could be examined and operationalized.

Important: This is not intended to be an exhaustive inventory of ethical concerns. It represents the concerns identified in the academic and industry evidence collected in this study.

2. From concerns to a structured solution

Classification using ethics labels. After consolidation, each concern cluster is mapped to the relevant cards and characteristics of the ethics-label framework proposed by Autili et al. The framework was published and validated with experts from academia and industry, and provides a practical way to communicate ethical dimensions of digital systems. In our study, the label acts as a semantic lens: it helps explain which ethical dimensions are affected and supports the transition from concerns to actionable recommendations.

Study logic at a glance



The ethics label provides a bridge between the evidence-based concern catalogue and practical use in software engineering.

What the proposed catalogue contains

A compact cycle of operational actions for each concern cluster

Operationalization. For each concern cluster, the proposed solution defines actions organized around the recurring cycle **Know → Protect → Inform → Assess → Verify → Respond**. Each action states an intended outcome and may include possible evidence/artifacts, SDLC phases, and responsible roles. These supporting elements are examples, not mandatory organizational formats.

Know Understand the concern, context, affected elements, and responsibilities.	**Protect** Apply proportionate safeguards and preventive measures.	**Inform** Communicate relevant information, choices, and limitations.
Assess Evaluate risks, impacts, trade-offs, and contextual conditions.	**Verify** Test, monitor, and check whether the intended controls work.	**Respond** Address adverse effects, remediate, report, and learn.

What you will be asked to do

- Review only the concern clusters assigned to you in detail.
- Assess whether the concerns are appropriately grouped and whether the mapping to the ethics-label framework is coherent.
- Assess the proposed actions for relevance, clarity, completeness, non-redundancy, and practical applicability.
- Identify gaps, overlaps, unclear elements, unrealistic assumptions, or practical limitations.
- After the assigned clusters, assess the catalogue as a whole and provide a short final reflection.

Participant instruction

There are no expected answers. We are particularly interested in disagreement, missing elements, overlaps, unclear wording, and practical limitations. Please evaluate the proposed solution on its merits rather than against the current practices of your organization.

How to read the worksheets

The upper section shows the concern cluster and its mapping to the ethics label. The lower section presents the six operational actions and the supporting examples. Reviewer fields are provided for your assessment and comments. Appendices A and B provide the complete concern-cluster catalogue and a concise ethics-label reference for consultation during validation.

Ethics-label framework: Autili et al. (2026), "Ethics label for digital systems to promote transparency and user awareness," The Journal of Systems & Software, 235, 112752.

Appendix A — Complete Concern-Cluster Catalogue

Reference material for assessing completeness, boundaries, overlap, and grouping. The catalogue reflects the 94 concerns collected from the industry survey and academic systematic mapping study; it is not intended to be an exhaustive taxonomy of ethical concerns.

ID	Concern Class	Primary Concerns
C-01	Privacy, Data Protection & Confidentiality	Privacy; Data Protection; Anonymity; Confidentiality; User Data Collection
C-02	Speech Issues	Speech Issues
C-03	Axiology	Axiology
C-04	Reliability, Dependability & Resilience	Reliability; Dependability; Resilience
C-05	Context, Individual Differences & Conflict of Interest	Individual Differences; Conflict of Interest; Context
C-06	Fairness & Equality	Fairness; Equality; Bias; Justice; Non-Discrimination; Inequality; Equity
C-07	Commerce & Monetization	Commerce; Monetization
C-08	Transparency & Traceability	Transparency; Traceability; Understandability; Observability; Openness; Lack of Data; Communication
C-09	Diversity, Inclusion, Plurality, Inclusiveness, Accessibility, Access & Cultural Sensitivity	Diversity; Inclusion; Plurality; Inclusiveness; Accessibility; Access; Cultural Sensitivity
C-10	Security & Malicious Use	Security; Encryption; User Data Collection; Malicious Use; Computer Abuse
C-11	Trust & Trustworthiness	Trust; Trustworthiness
C-12	Intellectual Properties, Authorship & Software Freedom	Intellectual Properties; Authorship; Software Freedom
C-13	Unemployment & Social Justice Issues	Unemployment; Social Justice Issues
C-14	Well-Being, Beneficence, Humanity, Care, Solidarity, Quality of Life, Common Goods & Retention/Addiction	Well-Being/Welfare; Beneficence; Humanity; Care; Solidarity; Quality of Life; Common goods; Retention and Addiction
C-15	Safety & Protection	Safety; Protection; Prevention of Harm/Do no Harm; Non-Maleficence
C-16	Technical Robustness & Data Quality	Controlled Failure; Technical Robustness; Data Quality; Accuracy; Consistency; Resilience; Lack of Data; Integrity
C-17	Explainability & Explicability	Explainability; Explicability; Communication; Interpretability
C-18	Compliance, Data Governance & Regulatory Approaches	Compliance; Data Governance; Regulatory Approaches
C-19	Professional Ethics & Competence	Professional Ethics; Work Ethics; Business Ethics; Competence
C-20	Utility, Usefulness & Usability	Utility; Usefulness; Usability
C-21	Freedom & Autonomy	Freedom; Human Agency; Respect for Human Autonomy; Autonomy; Dignity
C-22	Accountability & Responsibility	Accountability; Liability; Responsibility; Contestability; Oversight
C-23	Sustainability & Environmental Impact	Sustainability; Environmental Impact
C-24	Awareness & Communication	Awareness; Informed Consent; Self-Conception; Communication; Participation

Appendix B — Ethics-Label Reference

Concise reference for validating the mapping from concern clusters to the ethics-label framework. Terminology follows Autili et al.; descriptions below are shortened for reviewer use during validation.

Ethics Label	Characteristic	Short reference meaning
Human	Principle of Autonomy	Human control and ability to act according to informed beliefs, including intervention or regaining control when appropriate.
Human	Privacy and Data Governance / Intellectual Property	Privacy, personal-data governance, access/control, transparency of data management, user rights, and intellectual-property protection.
Human	Transparency and Explainability	Transparency of system use and outcomes, including understandable explanations of decisions.
Human	Beneficialness	Extent to which system use produces benefits for humans and supports well-being.
Human	(Not) Harmfulness	Whether system use avoids or may cause harm, including foreseeable misuse.
Societal	Society and Democracy	Effects on democratic processes, political engagement, deliberation, media pluralism, and information access.
Societal	Inclusive and Participatory Design	Inclusive design and involvement of relevant stakeholders and diverse user groups in development.
Societal	Openness and Plurality / Cultural Sensitivities	Openness, plurality of actors, local context, and recognition of cultural differences.
Societal	Diversity, Fairness, and Non-Discrimination	Fairness, diversity, inclusion, equality of opportunity, and avoidance of discriminatory or biased outcomes.
Societal	Responsibility and Accountability	Clear information about proper use, legal obligations, and who is responsible or accountable for system use and consequences.
Environmental	Energy Consumption	User-understandable information about energy used in system training and operation.
Environmental	Greenhouse Gas Emissions	User-understandable information about greenhouse-gas emissions associated with training and operation.
Environmental	Water Consumption	User-understandable information about water use during production/training and operation.
Environmental	Other Resource Consumption	Information about other material or natural-resource use, such as raw-material extraction and end-of-life disposal.
Environmental	Sustainability Promotion in Use	Whether system use supports or promotes more sustainable products, decisions, or recommendations.
Management of adverse effects	Minimization of Risk / Negative Impact	Plans and processes to identify and reduce human, societal, and environmental risks before or while they emerge.
Management of adverse effects	Mitigation of Risk / Negative Impact	Strategies to reduce the severity or consequences of identified or realized risks.
Management of adverse effects	Reporting Negative Impact	User-understandable reporting of potential or actual negative impacts and how they are addressed.

Reference: Autili et al. (2026), "Ethics label for digital systems to promote transparency and user awareness," The Journal of Systems & Software, 235, 112752. For full wording and examples, refer to the original publication.